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An Established Theory of Digital Twin Model for Tunnel Construction Safety Assessment

Yuhong Zhao [1,2], Naiqiang Wang [1,2] and Zhansheng Liu [1,2,*]

[1] College of Architecture, Civil and Transportation Engineering, Beijing University of Technology, Beijing 100124, China
[2] The Key Laboratory of Urban Security and Disaster Engineering of the Ministry of Education, Beijing University of Technology, Beijing 100124, China
* Correspondence: liuzhansheng@bjut.edu.cn

Abstract: In traditional construction safety assessment, it is difficult to describe the safety status of different construction stages. To solve this problem, this paper proposes a digital twin modeling theory for construction safety assessment. Firstly, this paper analyzes the requirements of a digital twin model. Secondly, the required information is collected by IoT. Finally, the DT model is established based on the collected information. This DT model analyzes the collected information by ML, which aims to conducting the assessments of construction safety. To verify this method, this paper analyzes the vault settlement during tunnel construction. The analysis results show that the DT model can predict the settlement value with high accuracy. Moreover, the safety state is assessed dynamically based on the settlement value by DT.

Keywords: tunnel construction; digital twin; machine learning; safety assessment

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1. Introduction

The construction of a tunnel is complicated and high-risk. In recent years, with the increase of underground projects, their construction process has caused more and more safety accidents to occur in construction. Due to the dangerous environment (such as unpredictable geological conditions) of tunnel construction, it is more dangerous in the tunnel construction process [1]. Therefore, it is of great significance to analyze and assess the safety status accurately in time, which can effectively avoid the risk of accidents and reduce accident losses. However, there is no real-time, comprehensive, and accurate method to analyze and assess the safety status of tunnel construction.

To obtain accurate safety assessment results, researchers have used various risk assessment methods to analyze the collected data about safety status. At present, quantitative methods such as analytic hierarchy process, support vector machine (SVM) and Bayesian network (BN) are mainly used. Zhou et al. [2] used BN to analyze the deflection risk of underground diaphragm walls to control the safety risk in the whole process of subway construction. Wei [3] combined the fault tree and analytic hierarchy process to evaluate and analyze the instability risk during tunnel construction, and the authors applied engineering cases to provide a decision-making basis for safety management. Taking the construction process of Qingdao Taishan railway station as the background, Liu et al. [4] combined the analytic hierarchy process (AHP) and expert classification method (EGM) to realize the marking of risk sources and risk control. Although the above studies made the analysis and assessment of the risks in tunnel construction more accurate, there are still some limitations. For example, the AHP calculation is very complicated. BN requires a large amount of training data; otherwise, it can easily lead to poor classification accuracy. Worse, most studies of safety risk assessment are limited to static risk assessments only, which are not suitable for the dynamic construction process of the tunnel.

In recent years, with the progress of science and technology, many countries have begun to promote the development of building industrialization and construction informatization, and monitoring technology also has made significant progress. More and more engineering data can be collected during the construction of tunnels. The engineering data used effectively is beneficial in controlling the safety risk in the construction process. Due to the characteristics of real-time interaction and self-evolution, digital twin (DT) is considered as the best technology that can contain all the required information [5]. Therefore, this paper uses DT to integrate and display all the construction data. This paper proposes a safety risk assessment framework with the DT model; the core of the DT model is all kinds of engineering data. Through establishing a digital twin construction model, various data in the construction process are analyzed and processed by machine learning algorithms. This paper introduces DT into the field of safety assessment and uses the characteristics of DT to establish a safety assessment method that meets the dynamic characteristics of construction. The safety assessment results are more consistent with the actual situation of the project and can better provide a basis for safety management. The digital twin model is based on the newly collected data for analysis and prediction to obtain the latest construction safety assessment. In brief, this work makes the following contributions:

- A construction safety assessment framework based on DT is proposed.
- The modeling accuracy of each construction element in the twin model is divided.
- A digital twin model including geometric, physical and knowledge dimensions is established. This model covers various elements of the construction project and provides a basis for various safety assessments.
- Taking tunnel structure safety evaluation as an example (vault settlement), the feasibility of this assessment method is verified.

The rest of this paper is as follows: the second chapter is a literature review, which introduces the characteristics and development of safety management and DT. The third chapter is about the construction safety evaluation framework of the DT. The fourth chapter introduces the modeling contents, information collection methods and modeling process in turn, including the establishment accuracy analysis of the virtual model, the appropriate method of data collection by the Internet of things (IoT), and the specific establishment process of the DT model. The fifth chapter verifies the above theory and framework through an actual tunnel construction case. The sixth chapter discusses the contribution and inadequacy of the article. The seventh chapter makes a final summary and outlook for the above content.

2. Literature Review

2.1. Risk Management

Risk management is the core component of safety status assessment. Einstein [6] is a leader in the development of risk management in tunnel construction. He used the method of risk analysis to study the construction process of mountain tunnels; Xiang W [7] et al. established a Bayesian network based on the existing case data and fault tree in the literature; the purpose of this study is to evaluate the risk probability of damage to existing facilities during the construction of underground structures. Ritter [8] et al. proposed a risk mechanism for existing buildings in shield tunneling in underground construction from the perspective that buildings are affected by people, mechanical equipment, construction materials, construction methods, construction environment, and other factors. Lyu [9] et al. present an improved trapezoidal fuzzy analytic hierarchy process (FAHP) to assess the risk related to land subsidence. Lin [10] et al. proposed a new risk identification model based on I-AHP and TOPSIS. The evaluated results provide a guide for adopting construction measures to mitigate risk and reduce accident occurrence. Zhou [11] et al. incorporated the original AHP and triangular fuzzy number-based AHP (TFN-AHP) into a geographic information system (GIS) to assess the inundation risk of the metro system in Shenzhen. The scholars mentioned above have played an essential role in promoting the development of risk management. They have improved the effectiveness and accuracy of risk management

to a certain extent. However, it remains difficult for risk management to meet the dynamic needs of the construction process, and feedback on the safety status of the construction site in real time is still the major unresolved problem. In recent years, information collection and transmission technologies such as the IoT have provided a foundation for solving the above problems. Shen [12] et al. proposed that dynamic risk management can be realized by combining the data collected by sensors with Buildings Information Model (BIM), machine learning and other technologies. This idea brings inspiration to this paper.

2.2. DT

The concept of DT was first proposed by Grieves in 2003. Grieves believes that DT is the digital information structure of a physical system, which contains all relevant information of its associated physical system [13]. DT establishes a multi-dimensional dynamic virtual model of physical entities to simulate and depict the attributes, behaviors, rules, etc. of physical entities in the real world. DT has a variety of definitions and interpretations, and there are many differences between the definitions and interpretations [14–16]. Nonetheless, DT interactive feedback and self-evolution characteristics have been widely approved [17,18]. DT was first used in aerospace and has now spread to all areas, including but not limited to manufacturing and construction [19,20]. Initially, DT was a three-dimensional architecture composed of physical entities, virtual digital models and interactive connections. Tao et al. [21] proposed the DT five-dimensional model by adding the concept of a twin database and service platform. In the manufacturing industry, DT is mainly used to describe the complex behavior of a system, considering the possible results of external factors, human-computer interaction and design constraints [22]. Qiu [23] proposed an AR assembly system architecture based on DT and analyzed the critical role of DT in the digital assembly process and the essential indispensable position of DT in the development of the digital state.

DT plays an essential role in the whole life cycle of the construction industry [24,25]. To enhance the adaptability of DT in the construction industry, many researchers have developed BIM as a medium technology for information interaction and display [26]. In the design stage, DT combined with BIM can provide designers with efficient real-time information [27]. Simulating a construction site based on BIM during construction, Zhao [28] et al. used DT to plan the path of the lifting process, which realized real-time interactive management of the lifting process and shortened the lifting route; Liu [29] et al. used DT to conduct safety management during steel structure construction, which realized effective prediction of safety status during construction. In addition, DT is widely applied to construction management and material monitoring [30,31]. DT has the most mature application in building operation and maintenance (O&M); Liu [32] et al. applied DT to the security field in the O&M, which improved the evacuation guidance effect. Moreover, the authors also have realized the intelligent planning of evacuation routes; Ye C [33] et al. applied DT to the field of structural health monitoring under building O&M and made a digital twin model for structural monitoring; Kim [34] et al. used DT to predict the life of the sound screen tunnel, which reduced the O&M cost; Shim [35] et al. proposed a detailed framework for DT to be used for bridge maintenance. DT can be used for fault diagnosis and maintenance decision-making, which helps improve decision-making efficiency and avoids data loss; Zhao [36] et al. proposed the fusion mechanism of the DT model and machine learning algorithm, which has promoted the practical application of the twin maintenance platform. DT collects data through IoT and realizes information interaction and data fusion using modeling technologies such as BIM. DT is crucial to improve intelligent safety management in future building construction.

2.3. Research Gap

Although DT has been applied and developed to a certain extent in the construction field, the research on risk management using DT is still relatively limited. There have been many research results in risk management, but it is still challenging to meet the need for the

construction process's safety risk assessment dynamically. With the rapid development of information collection technologies such as the IoT, construction personnel can quickly and accurately collect information. Therefore, this paper proposes a DT construction safety risk assessment framework that supports assessment dynamically. Through the establishment of the digital twin model, the data collected through various Internet of Things technologies will be fused and visualized. Through the knowledge model, all kinds of information data can be analyzed and processed, the feedback on safety status in the construction process can also be obtained, and the feedback can provide a basis for making safety management decisions in the construction process.

3. Construction Safety Assessment Framework Based on DT

Combined with the application of DT framework in other fields, a construction safety assessment framework is proposed [21]. This framework based on DT comprises three parts: an entity layer, digital layer, and function layer. The entity layer includes physical construction entities and sensor devices. The information, including man-machine, material, method, and environment are obtained by three-dimensional scanners and various sensor acquisition equipment. All the collected information is stored in the physical construction entities and can be synchronously mapped in the virtual construction model of the digital floor in real time. The digital layer is the core of the framework. The digital layer includes not only the virtual construction model that integrates all kinds of information data in the construction process but also the machine learning algorithm model. The machine learning algorithm model can be used to analyze the collected data and the simulation data obtained from the simulation of the virtual building model. The digital layer is a multi-dimensional model including a geometry model, physics model, and knowledge model. The functional layer can realize safety assessment and prediction based on the digital layer and the entity layer. To sum up, the entity layer is the foundation and data source for the establishment of the digital layer, the digital layer is the core for the realization of the functional layer, and the functional layer is the means for safety management. The DT construction safety assessment framework is shown in Figure 1.

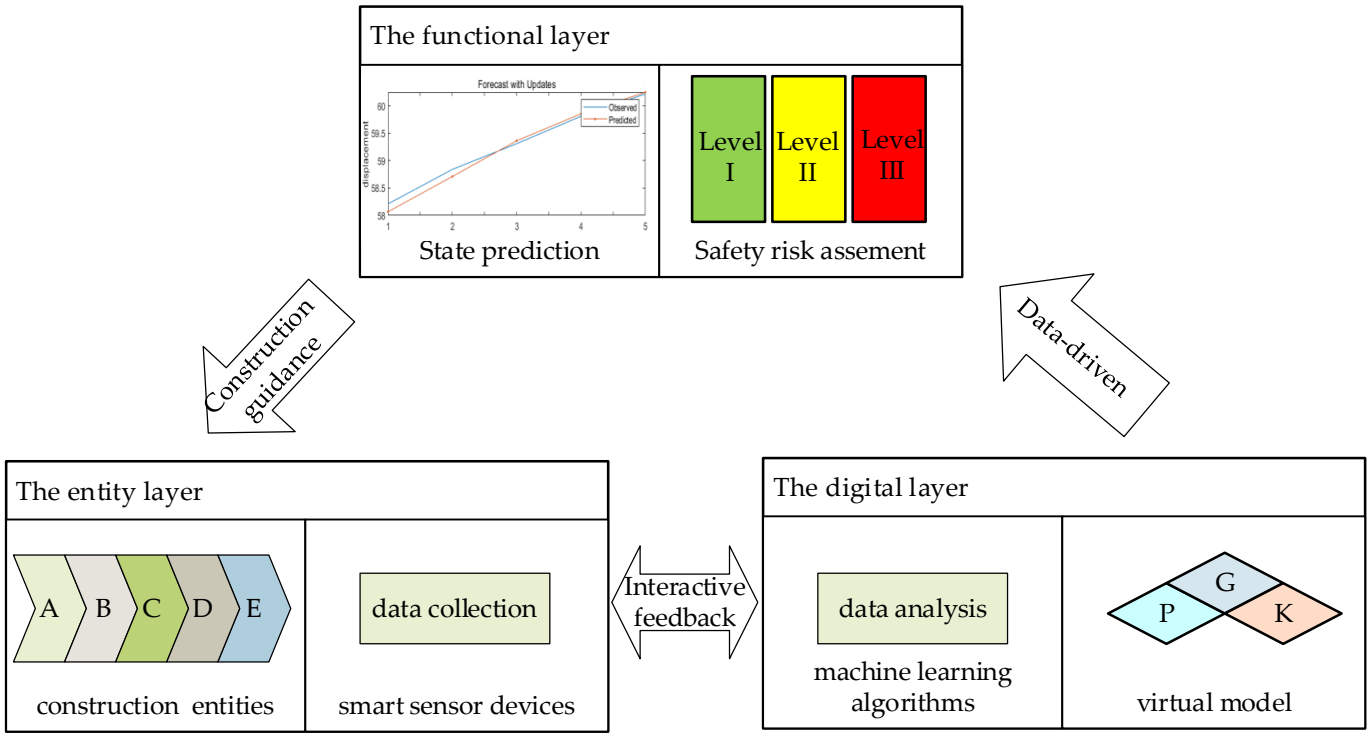


Figure 1. Construction safety risk assessment framework.

In Figure 1, A represents Men, B represents Machines, C represents Materials, D represents Method, E represents Environment, P represents the Physical model, G represents the Geometric models, and K represents the Knowledge model.

The expression of the digital twin model is as follows:

$$M_{CDT} = (M_{CP}, M_{CV}, M_{CN}) \quad (1)$$

$$M_{CV} = (M_G, M_P, M_K) \quad (2)$$

In formula, M_{CDT} is a digital twin model for construction safety assessment, M_{CP} is a physical construction site, M_{CN} is the connection between two parts, M_{CV} is a virtual construction model, M_G is a geometric model, and M_P is a physical model.

4. Establishment of DT Construction Model

According to the characteristics of the tunnel construction process and digital twin technology, a construction safety assessment model based on DT is established. The DT model needs to obtain the physical information and virtual model of tunnels. Moreover, the DT model also needs to fuse the collected information and virtual models. The establishment of the virtual model is mainly composed of three phases: definition coverage and a specific establishment method. Establishing the DT construction safety risk assessment model completes the data collection required, safety status analysis and analysis methods establishment.

4.1. Analysis of Content Covered by Virtual Model

The construction process of tunnels involves five major factors: men, machines, materials, methods, and environment, each element containing an extensive range of indicators and a variety of data. Therefore, the tunnel construction process is very complex. To establish a digital twin virtual model that meets safety assessment needs in the construction process, on the one hand the authors should avoid the inefficient problems caused by large data, and on the other hand the digital twin virtual model should include enough construction information to ensure the accuracy of the safety assessment results. To meet the above requirements, this paper determines each modeling object and its model level according to the order of modeling object accuracy level and dynamically changing element type. The steps of this process are shown in Figure 2 below.

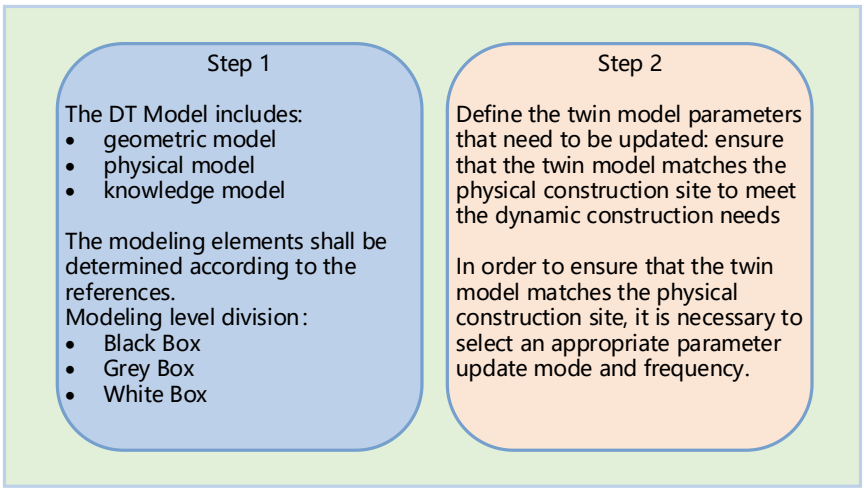


Figure 2. Two phases of creating the digital twin model.

4.1.1. Determine the Modeling Content and Its Accuracy Level

At this stage, the digital twin model and its modeling levels are determined according to the influence of different elements in the construction process on construction safety. The definition and classification of modeling accuracy are conducted through the hierarchical

method of the black box, gray box, and white box. Among them, the white box indicates the construction elements that need to be described as completely as possible; the gray box indicates the construction elements that need to be or can be only partially described; the black box shows the features that can only be described. Finally, the knowledge model is established through modeling behavior and rule dimensions, which endows the digital twin model with the ability to simulate dynamic construction behavior. The specific process can be divided into the following:

1. Establish coverage of construction elements that must be modeled. Experts' experience can be used to analyze and determine the construction elements that significantly impact the construction safety assessment. To reduce effort and digital models' complexity, the work should avoid modeling non-impact or low-impact construction elements. The premise is that the model can complete the simulation and simulation of the construction process.
2. Define the modeling level for each construction element. The modeling level should focus on the importance of construction elements, the comprehensiveness of available information, and the data characteristics of construction elements. Ideally, all construction elements should be defined to white-box precision. However, for some construction elements, not all the information data significantly influences the construction process's safety management. Some component information is incomplete, so it can be established at the gray-box level. The data, which has little influence on construction safety, can be established at the black-box level. It is worth noting that the data at the black-box level are still indispensable in ensuring the integrity of the construction twin model.
3. Finally, according to the construction process of the actual project, the virtual models of each construction element are integrated to obtain a complete digital twin model of the construction process. Based on the establishment of the knowledge-dimensional twin model, the twin model is given the ability and rules to evolve to be mapped and matched with the actual construction process in real time.

The specific steps are shown in the Figure 3.

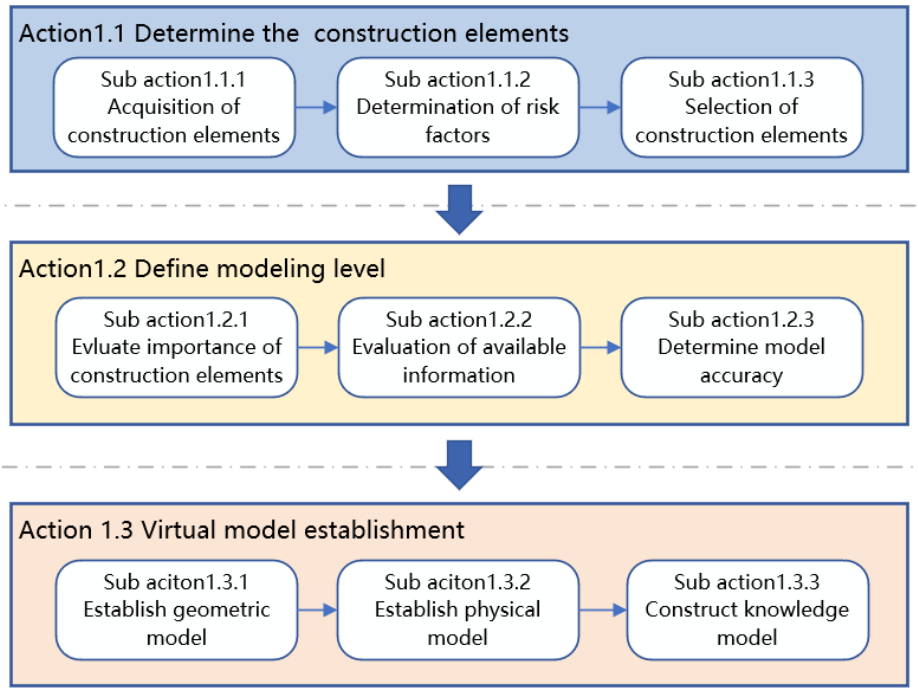


Figure 3. Actions and sub-actions of phase 1.

4.1.2. Define the Model Parameters That Need to Be Updated

The goal of this stage is to define the model parameters that need to be updated. The construction process is dynamic. To ensure the twin models can match the actual construction process well, it is necessary to update the model parameters according to the data collected by the sensors and the evaluation results obtained from the virtual model. The twin models need to be adjusted and optimized.

1. The completed twin model must be tested and debugged to ensure reliable real-time mapping of the actual construction process. The utility requirement of the overall model is a consideration that cannot be ignored when selecting model elements that need to be tuned. The adjustment of this process is mainly aimed at the knowledge model and related machine learning algorithms
2. Determining the relevant data for tuning is a critical task. When determining whether the data is needed or not in the tuning process, it must be considered whether this type of data can be obtained by a monitoring system or other reliable methods.
3. After identifying the twin models that need to be tuned and the data types required for tuning, selecting model parameters that can be updated on demand is the final step. The model adjustment work is achieved by adjusting the model parameters.

The specific steps are shown in the Figure 4.

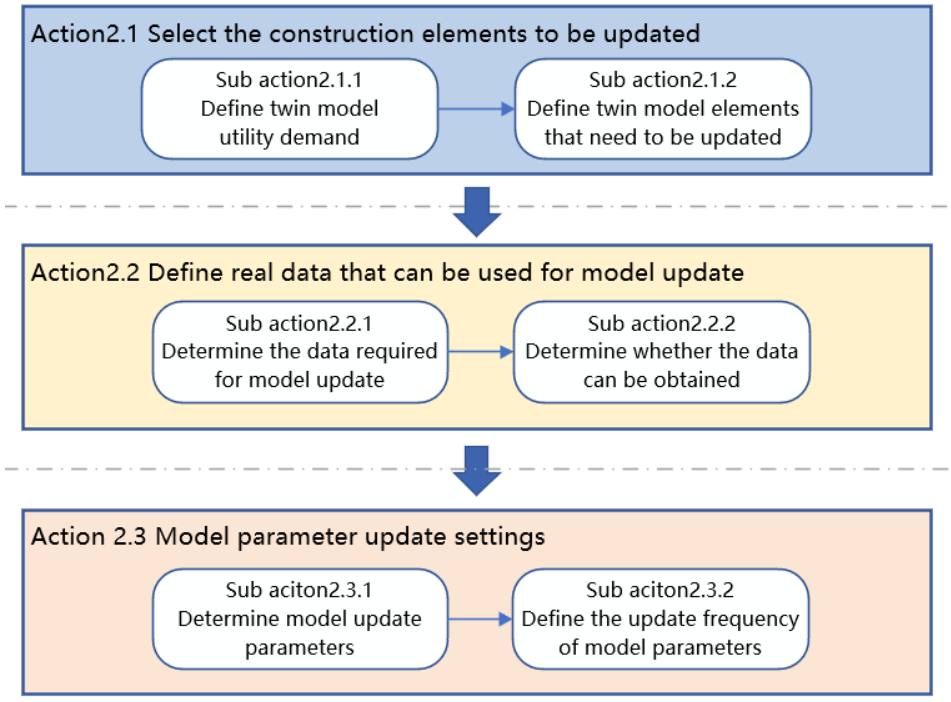


Figure 4. Actions and sub-actions of phase 2.

4.2. Physical Space Information Collection Based on Internet of Things Technology

To efficiently and accurately collect all kinds of data in the construction process, this paper proposes a construction process information collection system based on the IoT, which consists of a perception layer, a network layer, and a model layer. The physical space information collection system is based on the IoT. This system is the nervous system that connects the virtual world and the natural world to ensure the virtual twin model can be continuously updated to reflect changes in the real world (such as position, distance, speed, acceleration, direction, etc.). Data information, and data analysis, thereby guide the safety management of construction processes in the physical world.

The perception layer comprises a Radio Frequency Identification System (RFID), 3D scanners, various sensors, and other technologies. Once the construction starts, the

perception layer begins to collect multiple data, and different technologies can realize real-time monitoring of humans, machines, material, method, environments, and other factors. The perception layer realizes the conventional perception of the whole construction process, which is the basis of the information collection system.

The transport layer is the embodiment of the system data transmission. It realizes the real-time interaction between twin models and physical entities. There are two-way transmissions of data information as the path. Data transmission can be implemented by a combination of wired and wireless. This method ensures the data can be quickly, stably, and safely transmitted through a two-way route between the perception layer and the model layer. Data transmission is achieved through IoT communication technologies such as LoRa, Wi-Fi, Zigbee, Bluetooth, and 5G. Considering the characteristics of tunnel construction, it is recommended to use LoRa as the main transmission technology.

The information acquisition of the physical space provides the basis for establishing the virtual model and the evaluation of the safety state of the construction process, by arranging sensors and other monitoring instruments in the physical space to complete the information collection and then transmitting the data to the virtual model side through the network layer, thus laying the foundation for the establishment and updating of the model. On this basis, the process of simulation is established, which realizes analog synchronization between the digital world and the physical world during construction. The simulation is carried out through the virtual model to analyze and predict the construction process's safety state. The information collection process based on the IoT is shown in Figure 5.

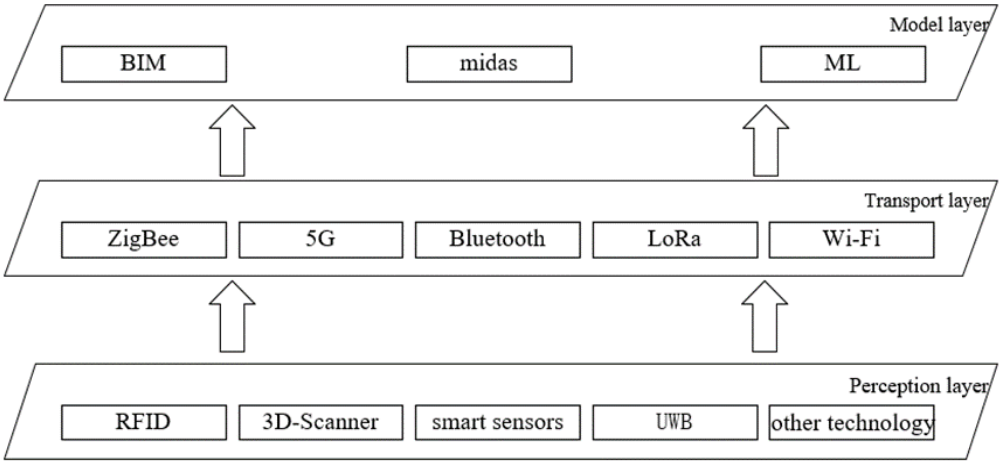


Figure 5. Information collection process based on the IoT.

As mentioned in Section 4.1 above, the tunnel construction process involves much information and data, which should be screened and sorted out during the modeling process. In this section, all the related information will be processed in mathematical expressions from the perspectives of men, machines, materials, and the environment.

The status information of construction personnel mainly includes location and health information. The mathematical expression to obtain the status information of construction workers is as follows:

$$PI_i = (ID_i, LI_i, HS_i) \tag{3}$$

In the above formula, PI_i represents personal information; ID_i represents personal identity information, which includes the information of name, job number, position type of job, etc.; LI_i represents the location information of personnel; and HS_i represents the health information of personnel, including heart rate, etc. All the information is obtained by intelligent helmets.

All sorts of sensors for information collection in the perception layer and construction equipment are called intelligent equipment. The mathematical expression to the information of construction equipment is as follows:

$$CE_i = (BI_i, WS_i, WD_i) \quad (4)$$

In the above equation, CE_i represents construction equipment information; BI_i represents basic information about construction equipment, including the model, function, and key parameters of construction equipment, etc.; WS_i represents the working status of construction equipment, such as working, idle, fault, etc.; and WD_i represents the relevant data during the working status of construction equipment, such as structural deformation data collected by sensors.

To establish an accurate and effective digital twin model, it is essential to obtain information on various engineering materials. The relevant materials' information is mainly divided into material names, material parameters, construction information, and other information. The mathematical expression for obtaining construction material information is as follows:

$$CM_i = (MN_i, MP_i, CI_i) \quad (5)$$

In the above formula, CM_i represents construction material information; MN_i stands for material name; MP_i represents material parameters, such as strength grade, elastic modulus, geometric dimension, etc.; and CI_i represents construction information, that is, the construction procedure and material function involved in the material.

In combination with the environmental characteristics during tunnel construction, geological data, harmful gas concentration and temperature information are the information to collect to reflect the construction environment. Therefore, the mathematical expression of environmental information monitoring and collection is as follows:

$$EI_i = (GI_i, PG_i, TM_i) \quad (6)$$

In the above formula, EI_i represents environmental information, GI_i represents geological information, PG_i represents the concentration of harmful gas, and TM_i represents temperature and humidity.

4.3. Establish Virtual Model

4.3.1. Data Integration Model

In the process of establishing the DT model, the twin model is often divided into four levels: "geometry model, physics model, behavior model and rules model" [37]. Through the information collection of actual objects in the physical world, the information collected and the model established are completed. The multi-level and high-precision simulation of the actual construction process is realized by associating and integrating various dimensional models.

BIM has been widely used in construction in recent years. BIM has the advantages of high precision, complete information and multi-dimensional visualization, which can be very competent for the establishment of a geometric dimension model of twin models. Establishing geometric models by BIM can realize the data visualization of the construction process, which lays a foundation for the establishment of other dimension models. BIM modeling software such as Revit can be selected to complete the establishment of the geometric model. The high-precision BIM model can reflect the geometric characteristics of the construction process, and it can be a suitable carrier of the basic information of various construction materials and components. However, due to the complexity and variability of the actual construction process, it is difficult to ensure that the actual construction process is entirely consistent with the BIM model [38]. To reduce the error between the BIM model and the actual construction process, 3D scanning technology is used to capture the actual construction state. The 3D scanning needs to obtain the point cloud data model of the

construction process firstly, then establish the point cloud model and integrate the point cloud with the BIM model. Combining the point cloud model and BIM model can adjust the geometric model in time, which improves the accuracy and precision of the geometric model [39]. After the point cloud data is obtained, the abnormal data caused by machine error and external factors in the scanning process are deleted by denoising. The adjusted BIM model can more accurately reflect the geometric characteristics of the construction process and provide a basis for the adjustment of the physical model, thus improving the accuracy of the physical model. At the physical level, the physical characteristics of construction objects are mainly simulated through finite element analysis software such as MIDAS. The construction object's mechanical performance is affected and calculated on this basis of finite element analysis, combined with the sensor equipment's data. In the actual construction process, the geometric dimensions of the building components impact the mechanical properties. Therefore, high-precision geometric models can improve the finite element analysis' accuracy and the reliability of physical models.

The behavior model proposed by Tao Fei is mainly used to describe the real-time response and behavior of entity objects under the set operation mechanism, such as evolutionary behavior, performance degradation behavior, etc., which is commonly achieved by machine learning [18,37]. The rule model is established and combines the behavior rules of entity objects with relevant standards and specifications [18,37]. The purpose of the rule model is to evaluate and predict the performance of entity objects in the future. Generally, it is generated by integrating existing knowledge and machine learning algorithms to continuously evolve and mine.

As mentioned above, the behavior model and rule model have certain similarities and close connections. The rule model provides guidance for the behavior model, and the behavior model and rule model are both inseparable from machine learning algorithms. Therefore, this paper integrates the two models into one model through the Data-to-Information-to-Knowledge-to-Wisdom Model (DIKW) hierarchy theory. The integrated model is called the knowledge model. DIKW refers to the transformation and improvement process of "data $\rightarrow$ information $\rightarrow$ knowledge $\rightarrow$ wisdom" [40]. This paper uses the data collected by sensors and simulation data as "Data". Information is obtained after data filtering and association. The "information" means associated data that point to the application target. Next, the information is analyzed and mined by the machine learning algorithm, which aims to find the rules of data evolution. Then, the rules of data evolution are endowed into the virtual model with the ability of behavior evolution, that is, "knowledge". When the "knowledge" is obtained, the model has the functions of judgment, evaluation, prediction, etc. In other words, the model can provide decision support for project managers after obtaining "knowledge". The "knowledge" gives the twin model the ability to find and solve problems, which means the twin model is "Wisdom". In the process of building the knowledge model, the machine learning algorithm plays an important role. See Section 4.3.2 for details.

Figure 6 shows the process of the establishment of the twin virtual model.

4.3.2. Data-Driven Machine Learning Models

In the construction process, construction elements change with time. To make the virtual model keep a real-time reflection of the real world, machine learning algorithms are introduced into the knowledge model to analyze and process the data. Machine learning algorithms give the virtual model the ability to interact and evolve with the physical world by mining data rules.

The long-short term memory (LSTM) and other time-space series prediction algorithms are used for prediction during construction [41]. Therefore, this paper chooses LSTM as the driving algorithm of the knowledge model.

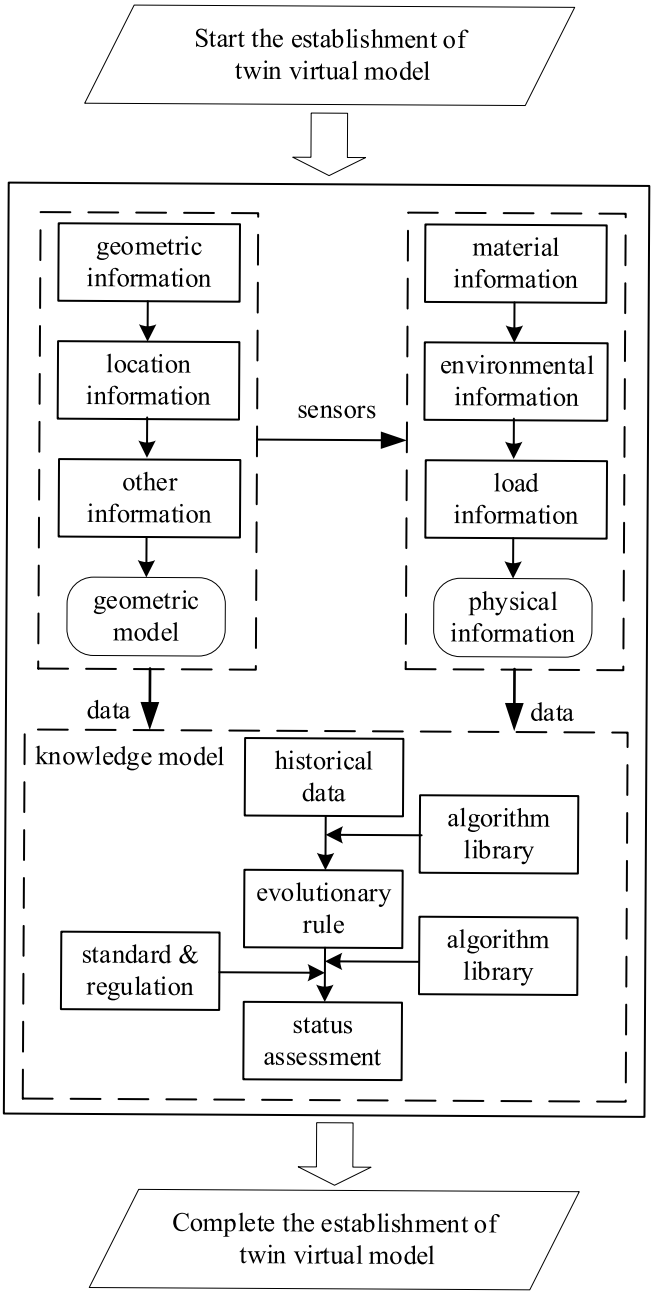


Figure 6. The process of the establishment of a twin virtual model.

LSTM makes up for the defect of the Recurrent Neural Network which finds it difficult to solve the problem of long-distance dependence. LSTM consists of an input layer, an LSTM layer, a fully connected layer, and an output layer. In the LSTM layer, the transmission of information is controlled by the input gate, forget gate and output gate inside the neuron. Each gate has an activation function, called $\sigma_1, \sigma_2, \sigma_3$. There is an output function in each input gate and the output gate. The specific structure is shown in Figure 7. In the figure, "pro" represents the multiplication relationship between elements, and "sum" represents the addition relationship between elements. f_t represents the processing result of the activation function of the forgetting gate, i_t represents the processing result of the activation function of the input gate, and o_t represents the result of the activation function processing of the output gate. Unit status is the core element of LSTM, and C_{t-1} is the memory information output from the previous unit, which is processed by the function of the current LSTM layer to obtain the output memory information C_t. h_{t-1} and h_t respectively

identify the output results of the two LSTM units, and X represents input information. The specific relationship is as follows:

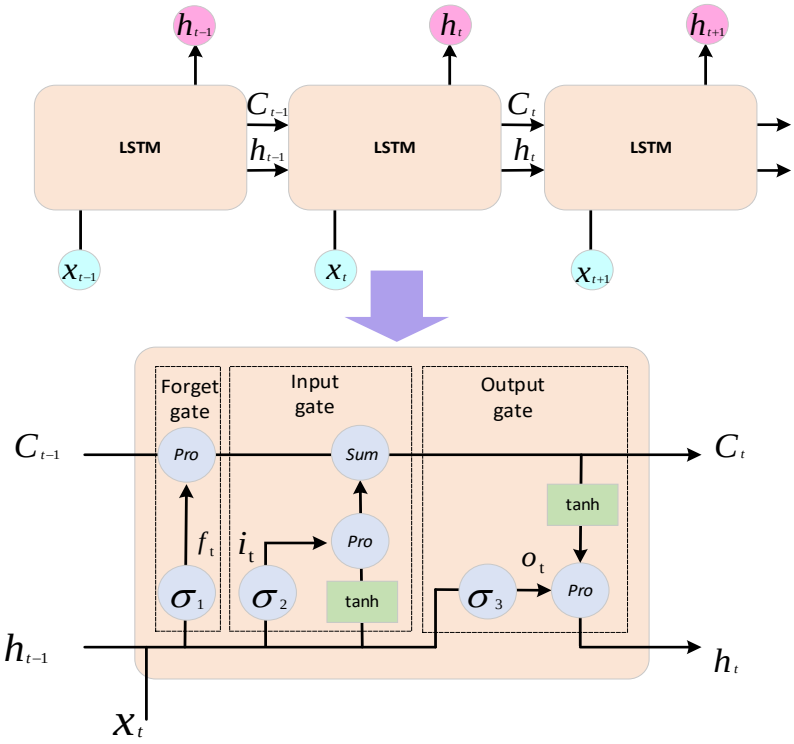


Figure 7. The structure of LSTM.

The specific process of LSTM operation is as follows:

1. Input and standardize data. The data loading algorithm is divided into the training set and test set, which are divided into 90% and 10% in this paper. The data are standardized to improve the efficiency and prediction accuracy of the algorithm, which can obtain better fitting and prevent training divergence.
2. Algorithm parameter setting. In this paper, the time window is positioned as a time step; at each time step of the input sequence, the LSTM network learns to predict the value of the next time step. The parameters of the prediction network are set, including training times, gradient threshold, initial learning rate, learning rate decline cycle, learning rate decline factor, etc.
3. Algorithm training and prediction effect check. In this paper, after the above parameters are set, the algorithm model is trained. If the training result is qualified, input the test group data to verify the prediction effect. If the training result is unqualified, adjust the above parameters and repeat the above steps.

5. Case Study

5.1. Modeling Accuracy and Content

According to the rules described in Section 4.1.1, the construction elements of a tunnel construction process are classified and analyzed according to men, machines, materials and environment. What's more, the black, gray and white levels are classified according to their different impacts on construction safety and available data. In this study, RFID tags are used as the main technology to store information of various construction elements. The information stored in RFID tags can be updated and edited in real time. Therefore, we can choose the appropriate time point to change the status information of construction elements. Before construction, the specific information of construction elements is gathered by scanning the electronic tags on the components. All kinds of structural information are monitored by various sensors, and the data collection interval is two seconds.

In the construction process, human safety is a key consideration in safety management, and human status has an important impact on the overall safety of the construction. Therefore, it is necessary to collect information that is as detailed as possible on construction personnel, including age, type of work, working years, working status, location, etc. However, because geometric information of construction personnel has little impact on the construction structure, there is no need to establish a geometric model; consequently, the model accuracy of construction personnel is at the gray-box level. The model expression of construction personnel is as follows: PI_1 = {{SG001, M, Constructor, 35, 10}, {228.2, 202.3, −11}, 78}. The specific meaning is {{Job number, male, type of work, age, working years}, location coordinate, heart rate}. The coordinate origin of the location coordinate is a point on the site elevation of the Project Department of the construction bid section.

The status of construction machinery and equipment has an essential impact on the progress and quality of the construction project. Therefore, it is necessary to ensure that the construction machinery and equipment maintain normal working conditions. However, since much construction equipment does not belong to the project itself, it is not necessary to build a geometric model, and the gray-box level is sufficient.

Manage relevant data on construction equipment during construction. As described in Section 4.2 above, RFID is filled with pertinent information on mechanical equipment. Firstly, input the mechanical equipment's basic information and working status information into the RFID tag and paste RFID tags onto the corresponding equipment. Before construction, the RFID handheld terminal can scan the electronic tags on the equipment to learn the functionality of mechanical equipment. Meanwhile, the information for the working status of the equipment can be changed from "idle" to "working". Taking the segment installation machine during shield tunneling as an example, its model expression formula is CE_1 = {{CREC010, segment installation machine, Segment assembly, 120 kN@130 bar, 320 kNm@210 bar, 400 kNm@260 bar, 6 degrees of freedom, ring gear type, mechanical gripping, Hydraulic drive}, working, {28 °C, 580 MPa, 300 MPa, 0.3 m, 0.7 rpm}}. The specific meaning is {{ID code corresponding to the machine model, equipment type, function, rated lifting capacity, rotating torque, static torque, type, driving mode, and other core parameters}, operating state of the equipment in working, {operating temperature, contact stress, gear bending stress, displacement, rotation speed}}.

The materials are the basis of the construction structure. In this paper, the structural components are the basic units of construction material modeling. The BIM model containing the construction information such as the material type, strength, and location is established as the constituent elements of the geometric model of the construction structure. Therefore, the modeling level of materials should be at white-box level.

In this study, the RFID tag is also used as the main technology to store information on various construction materials. Taking a segment as an example, its model expression formula is CM_1 = {AT334190314T, {segment, 6000 mm, 5500 mm, 1000 mm, C50, P10, HPB300, HRB400}, reinforced concrete, shield interval 1, 2018-04-12 of a construction section of a project, segment factory of a construction bureau of a city, {shield normal tunneling-segment assembly, 15 min, qualified}, {232.2, 200.7, −11}}. The specific meaning is {The unique identification ID corresponding to the segment and the active RFID, {material type, outer diameter, inner diameter, ring width, concrete strength grade, impermeability grade, reinforcement type, reinforcement grade}, material, location, production time, production unit, {the construction process in which the segment is located, the time spent in the process, quality grade}, {location coordinate, (unit: meter)}}.

As an external factor that may impact construction safety, environmental information may lead to a more severe loss of life and property. Therefore, environmental information is a construction element that cannot be ignored. In this paper, the environmental factors that may affect construction safety are divided into two categories. One is the environmental factors that directly affect the structural safety, represented by the surrounding rock. The other is the environmental factors that affect the workers' status (such as the concentration of harmful gases); among them, the surrounding rock and other factors are mainly

considered for their mechanical properties, which are defined as the white-box level. The concentration of harmful gases and other relevant environmental factors only need to be considered as simple numerical information, which is defined as the black-box level.

The surrounding rock information can be stored in the digital twin model by the construction of geological exploration results. The model expression formula is GI_1 = {{0.108, 0.125, 5.84}, 13.62}. The specific meaning is {{rock mass integrity coefficient, surrounding rock quality index, compression modulus}, surrounding rock deformation value (unit: mm)}. The model expression formula of harmful gas concentration and other related environmental factors is EI_1 = {0.01%, 18 °C, 0.4% rh, 3 m/s}. The specific meaning is EI_1 = {concentration of harmful gas, temperature, humidity, wind speed}.

The tunnel construction process is dynamic, and many of the construction elements are not static. Many of the above elements should be monitored and updated with appropriate frequency. The purpose of this paper is to evaluate the construction safety state based on the digital twin model. The settlement, which can represent the main factor of construction structure safety, is selected as the critical monitoring and analysis parameter [42]. According to the requirements of relevant standards, the data collection frequency is once a day. The collection results are both the basis for model updating and essential data for risk prediction.

5.2. Virtual Model Establishment

Based on the construction information obtained by various information collection devices such as intelligent sensors and RFID, the digital twin virtual model of the tunnel construction process is established to realize the "geometry physics knowledge" data integration.

The basic geometric information of the tunnel structure is obtained through CAD graphics, and the initial geometric model is established using BIM technology. On this basis, the Trimble TX5 3D scanner is used to conduct 3D scanning of the construction structure, to achieve the correction of the geometric model. This modeling method ensures that the geometric model can truly reflect the geometric information of the construction structure. Based on the established BIM geometric model and the collected construction information, the basic information of various components is supplemented. The MIDAS model was established by Revit Api from the geometric model [43]. Combined with the sensor collection information, the same working conditions are set as in the actual construction process, and the physical model is established. The mechanical property of the construction structure is analyzed, and the mechanical property parameters obtained can be directly used for the structural safety assessment, which provides a basis for the establishment of the knowledge model.

The establishment of a geometric model and physical model is the application process of modeling software, which is not described in this paper. The practical application of algorithms in the knowledge model is introduced in detail. Based on the construction information provided by the geometric model and physical model and structural design specifications, a rule database is established. The data analysis of the twin model is conducted using the LSTM, and a knowledge model is established to endow the twin virtual models with the ability of self-evolution, to realize the simulation, analysis, and evaluation of the tunnel construction process.

When the physical information acquisition and data integration model are completed, this paper selects the LSTM algorithm to analyze and process the construction data. According to the current situation and change trend of various risk factors, the prediction for the safety status of the construction process on the next construction day is realized. As mentioned above, this paper selects the settlement deformation of the tunnel as an example to analyze the safety state of the construction process. Through the knowledge model, the predicted settlement is compared with the relevant standards and regulations, and timely adjustment measures can be taken according to the settlement to ensure the overall safety of the construction process.

5.3. Predictive Performance Analysis

In this paper, a double-hole one-way tunnel is selected as an example. Taking a section of the tunnel as the research object, the building boundary of the tunnel is 10.25 m × 5 m, and the surrounding rock level of the monitoring section is Grade V. As described in Section 5.1, this article carries out structural safety assessment by analyzing the settlement value. During monitoring, measuring points are arranged at the arch crown, left arch and right arch respectively, and the average value of the three measuring points is used as the arch crown settlement value. According to the actual project overview and relevant specifications, the design limit displacement of the tunnel is 120 mm [44]. The displacement management level can be divided into three levels, and the specific rules are shown in Table 1 below:

Table 1. Displacement Management Level.

Level	Displacement Relation/mm	Security Status
I	$U < U_0/3$	normal
II	$U_0/3 \leq U \leq 2U_0/3$	reinforcement
III	$U > 2U_0/3$	special treatment

Note: U is the measured value of vault settlement; U_0 is the design limit displacement value of the settlement.

The collected settlement data is divided into a training set and a test set according to the proportion of 9:1. LSTM parameters are as follows: The max epoch is 300, the learning rate is 0.005, the gradient threshold is 1, the learning rate decline cycle is 150, and the learning rate decline factor is 0.2.

The effect of the test set is shown in the Figure 8. It can be seen from the figure that its RMES is 0.092924, the absolute value of the maximum error is about 0.135 mm, and its prediction effect and accuracy meet the requirements for judging the construction safety state.

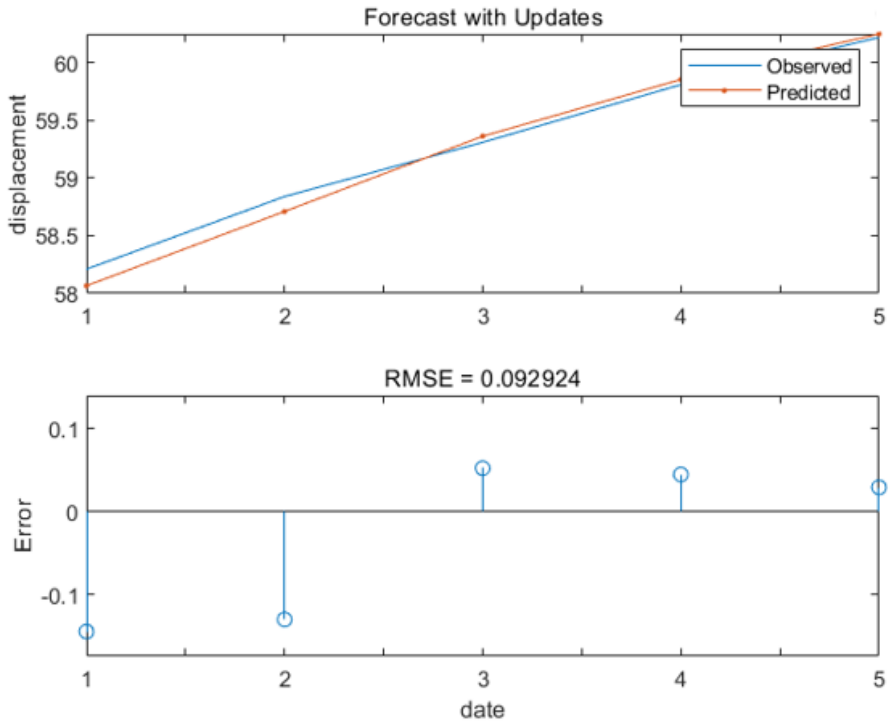


Figure 8. The LSTM prediction effect analysis chart.

By combining the prediction results with the relevant standards in the knowledge model, it is judged that the tunnel construction structure safety is always at level II, and no special operation other than reinforcement is required.

6. Discussion

In this paper, digital twin technology is applied to the field of safety management in the construction process, to make up for the deficiency that the traditional safety evaluation cannot meet the dynamic need of the construction process. By establishing a construction safety assessment framework based on digital twins, the basic relationship between digital twin technology and safety status evaluation is described. This paper aims to realize the data integration and visualization of various construction elements by establishing a twin virtual model, so as to provide a data basis for various safety evaluations. This paper analyzes the data content and accuracy classification that the twin model needs to cover, which lays a foundation for the subsequent modeling research. In this case, by observing the error value and RMSE of LSTM used for settlement prediction, it is concluded that the algorithm can be applied to structural safety assessment. Taking the structural safety assessment as an example, the feasibility of this safety assessment method in other safety fields during the construction process is inferred.

Next, researchers will continue to promote the implementation and application of this construction safety assessment. The researchers will verify the practicability of the evaluation method with as many kinds of security risks as possible, so as to enhance the persuasiveness of the theory.

7. Conclusions

In this paper, a digital twin model of tunnel construction for intelligent safety evaluation is proposed. Moreover, a construction safety evaluation framework and modeling method are described in detail. In this study, the real space information collection technology represented by IoT and by intelligent sensor technology is used to collect data from the actual project. The virtual twin model is established by combining BIM technology, Midas technology, and the machine learning algorithm represented by LSTM. The feasibility of this method is verified by taking the arch crown settlement during tunnel construction as an example. In this paper, the main conclusions are as follows:

- The information obtained by IoT and smart devices is the basis for establishing the digital twin model. The process and analysis of the information are the core of evaluating and predicting the construction safety status. In the face of a large number of miscellaneous construction data in the construction process, this paper uses the hierarchical method of the black box, gray box, and white box to process a large number of miscellaneous construction data. The method can reduce the workload of the virtual model building process and improve the efficiency of virtual model operation, and it meets the use requirements of the virtual twin model. It provides a reference for subsequent modeling research.
- By combining the relevant standards and specifications covered in the knowledge model, the safety evaluation and state prediction of each construction element in the construction process can be realized. The results of safety evaluation and state prediction are also the basis for the digital twin model to simulate the construction process and map the construction site in real time.

To sum up, this paper realizes the evaluation and prediction of construction elements in the construction process through DT, which meets the dynamic requirements of the construction process. Besides, the twin models realize the digital closed-loop of safety state evaluation and prediction, which ensures that the construction process is in a safe state all the time. The safety management in the tunnel construction process is more intelligent in this paper. This study provides a new idea for intelligent safety management in the construction process.

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